Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/25/2018
PI Name:	Liu, Bin Ph.D.		
Project Title:	Three-Dimensional Dusty Plasma Experiments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	07/25/2017	**End Date:**	04/25/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Goree, John Ph.D. (University of Iowa)		
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Task Description:	A flight experiment will be performed to study phenomena resulting from shear motion in a dusty plasma. Applying an external shear, we will probe fundamental material properties selected from these topics: lattice wave excitation and coupling, viscous transport, and microscopic motion under frictional conditions. We will take a microscopic approach to uncover the nature of macroscopic properties. We will use mesoscopic particles, which serve as analogs of molecules, but allow tracking of individual particles as they interact. The experiment will use dusty plasmas, also known as "complex plasmas." These are four-component mixtures of micron-size solid particles, electrons, ions, and neutral gas. The solid particles obtain large electric charges, so that they interact by Coulomb forces with neighboring particles. The charges are so large, that these particles behave collectively as a strongly-coupled plasma. The cloud of solid particles constitutes the sample that is studied. Microgravity enables our experiment with a three-dimensional sample. Without microgravity, under 1g conditions, sedimentation causes the dust cloud to settle into a 2D layer, or at most a few horizontal layers. Our microgravity experiment will extend the understanding of particle motion that was gained in previous ground-based experiments with smaller 2D samples. Our flight experiment will be performed using PK-4, which is a facility in the Columbus module of the International Space Station (ISS). It was built by the European Space Agency. The main diagnostic is video imaging using cameras. Image data will be analyzed to generate particle position and velocity. Physical measures that can be obtained from these image data, using methods developed for ground-based experiments, include a wave’s propagation speed and dispersion relation, temperature profile, flow velocity, or shear stress. SCIENTIFIC/TECHNICAL MERITS The experiment's goals are to answer at least one of these fundamental physics questions: (1) Can we exploit the unique properties of dusty plasma to verify the theoretical prediction of localized viscous heating in a 3D liquid? (2) Can hydrodynamic shear motion couple into the generation of longitudinal waves? (3) What is the dispersion relation of lattice waves in a liquid? (4) Does a three-dimensional dusty plasma exhibit stick-slip motion during plastic deformation? The experiment will exploit manipulation devices on PK-4 including laser manipulation. Laser light imparts momentum to the solid particles, causing them to flow through the surrounding sample region. This modulation will be a pulse that is either long or short, according to the scientific goal. The feasibility of our flight experiment is based on previous ground-based experiments and our simulation. Our laboratory experiments have confirmed localized viscous heating and lattice waves. Ground-based experiments by other scientists have also confirmed the existences of plastic flow conditions and stick-slip motion. Using a simulation under PK-4 conditions, we have established the physical mechanism of wave excitation by a pulsed shear flow. In the unlikely event that this flight experiment does not yield useful results, we have a plan for a backup experiment to investigate the superheating and supercooling phase transitions, or non-Gaussian statistics. BROADER IMPACTS This research has an overlap with statistical physics, fluid physics, and soft condensed matter physics, as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, and a website. A K-12 outreach presentation will be made at an elementary school.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has an overlap with statistical physics, fluid physics, and soft condensed matter physics, as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, and a website. A K-12 outreach presentation will be made at an elementary school.
	During this period, we carried out a fundamental physics research to prepare for the forthcoming flight experiments: Three-Dimensional (3D) Dusty Plasma Experiments, using the PK-4 instrument on the International Space Station (ISS). Our research activities included flight experiment planning, analysis method development, and computer simulation. Significant progress has been made in analyzing the PK-4 flight data from previous missions. We published our first peer-reviewed journal paper that uses the PK-4 flight data. To support our flight experiments, we also performed a computer simulation, yielding a second publication in a peer-reviewed journal. For our research group, during this report period we published five peer-reviewed journal papers, and presented two invited and seven contributed talks. This research focused on fundamental physics at the microscopic particle level, using strongly-coupled dusty plasmas. Small particles of solid matter, also called dust, are immersed in a mixture of electrons, ions, and neutral gas, making a dusty plasma. The solid particles obtain large electric charges, so that they interact by Coulomb forces with neighboring particles. The charges are so large that these particles behave collectively as a strongly-coupled plasma. The cloud of the solid particles constitutes the sample that is studied. Direct imaging of dust motion offers a unique opportunity to uncover the nature of particle behavior at the microscopic level. FUNDAMENTAL PHYSICS AT THE PARTICLE LEVEL Using direct imaging of dust motion, we can probe material fundamental properties selected from these topics: lattice wave excitation and coupling, viscous transport, and microscopic motion under frictional conditions. Our flight experiment's goals are to answer at least one of these fundamental physics questions: (1) Can we exploit the unique properties of dusty plasma to verify the theoretical prediction of localized viscous heating in a 3D liquid? (2) Can hydrodynamic shear motion couple into the generation of longitudinal waves? (3) What is the dispersion relation of lattice waves in a liquid? (4) Does a 3D dusty plasma exhibit stick-slip motion during plastic deformation? (5) Is there a superheated state? Can a 3D dusty plasma be supercooled? (6) Does the motion of particles deviate from Gaussian statistics, with distributions exhibiting kurtosis and skew? FLIGHT EXPERIMENT PLANNING

Task Progress:

Our planning involved a collaboration with other institutions or agencies. Significant collaborators included PK-4 Core Team members in Germany and Russia, PK-4 Facility Science Team (FST), ESA (European Space Agency) officials, and the collaborators specific to Iowa projects. Both Principal Investigator (PI) Bin Liu and Co-Investigator (Co-I) John Goree interacted with these collaborators frequently via emails and occasionally by phone, and we also met them at various meetings and conferences. One significant meeting was the recent Facility Science Team (FST) meeting at the DLR (German Space Agency) in December 2017. Both PI Bin Liu and Co-I John Goree attended this meeting.

Working with our collaborators, we made a general plan for our forthcoming 3D dusty plasma experiments using the PK-4 instrument onboard the ISS. A preliminary experiment is tentatively planned for late 2018, followed by another experiment about a year later.

To prepare for the forthcoming flight experiment, we performed an on-site laboratory test at the DLR. We did this test together with PK-4 Core Team scientists at the DLR, using a PK-4 Ground Model (the apparatus was the Scientific Reference Model at the DLR). In this test, we confirmed that the PK-4 manipulation laser can meet our scientific requirements. The most important outcome of this test was a first draft of the flow charts, which will be used to develop more detailed scripts for the upcoming Iowa flight experiments.

Planning activities on-site at Iowa included a survey of the flight data from previous missions. During the past year, the Iowa group received the flight data from four previous missions, and we have so far surveyed the data for three missions. To expedite our data analysis, we also hired two undergraduates, who were funded by National Science Foundation (NSF).

OBSERVATION OF NON-GAUSSIAN STATISTICS

We have so far achieved one of our scientific goals proposed for this project, on the study of statistical physics at the particle level. We can now answer the following question raised in our project proposal: Goal #6: Does the motion of particles deviate from Gaussian statistics, with distributions exhibiting kurtosis and skew? We found that the particle motion in a 3D dusty plasma under microgravity conditions deviated from Gaussian statistics, with a velocity distribution function exhibiting high-energy tails.

Our finding was based on an analysis of the PK-4 flight data from a previous mission. The data was obtained from the Science Campaign #1 of the PK-4 instrument onboard the ISS. We chose an experimental sequence where we could analyze a region with very little flow velocity. A 3D cloud of dust particles was trapped by a RF coil, in a neon plasma at 1 mbar. These dust particles were tracked using video microscopy in a cross section of the 3D dust cloud. Images of particle motion were captured at a frame rate of 35 Hz. We tracked particle motion frame to frame, yielding a velocity distribution function.

Our finding was published in a peer-reviewed journal. In our paper, we reported our experimental observation of non-Maxwellian velocity distribution function (VDF) with high-energy tails. In addition to this experimental finding, we further characterized the non-Gaussian distribution using a theoretical model. We found that the VDF is fit well by a combination of low-energy Maxwellian core and a high-energy non-Gaussian Kappa-distribution halo. This finding is surprising, because similar non-Maxwellian VDF has been observed previously in the solar wind, which is nonequilibrium and mostly collisionless. The dust particles in our plasma, on the other hand, are highly collisional among themselves.

In addition to the significant finding discussed above, another area of progress was the development of software for our flight data analysis. Using this software, we successfully extracted experimental parameters from PK-4 raw data products. This progress is significant because PK-4 raw housekeeping data products are not formatted in a way that is understandable to scientists. Therefore, we have developed our own computer code to extract the experimental parameters, and we have shared this software with other NASA-funded PIs.

Furthermore, our data survey helped us in optimizing flight experiment conditions. We searched for the experimental conditions to meet our need for a 3D dusty plasma flow, yielding a graph of particle speed vs laser power, when laser manipulation was used in PK-4. Using an analysis method called a “space-time diagram,” we identified the gas pressure range where plasma glow exhibits a significant disturbance called a striation. We expect that striations will be a common feature that will often affect PK-4 experiments. We also found a limitation in the early PK-4 flight data, due to the choice of camera-frame-rate parameters. This helped us in better defining our experimental requirements.

ANALYSIS METHOD DEVELOPMENT

To prepare for our future flight experiments, we developed an analysis method to characterize the shear motion, as can be induced by the PK-4 manipulation laser. Based on the parameters verified by our recent breadboard experiments on the shear viscosity, we devised a scheme to characterize the deformation of dusty plasma under shear stresses. We also performed a numerical simulation to validate this scheme. Using this scheme, we successfully obtained shear strain, shear stress, and a stress-strain relationship curve.

COMPUTER SIMULATION

Our computer simulation was to test physics concepts, assess the feasibility of our future PK-4 experiments, and develop analysis methods.

We performed a simulation on the deformation under shear stresses. In this simulation, we tested interdisciplinary concepts in rheology: shear stress, shear strain, and stress-strain relationship. We also assessed the feasibility to carry out flight experiments to study plastic deformation of a solid-like cloud of dust particles, where the deformation is the result of manipulation laser beams. The simulation mimicked experiments that start with a crystal that was then manipulated by a pair of laser beams separated by a gap. In a pair of rectangular regions, we applied two equal but oppositely directed forces, to induce a shear deformation in the gap between. These external forces were increased incrementally to examine the elastic behavior, plasticity, and liquid flow.

Our simulation findings on the deformation was published in Physics of Plasma. The title of our paper is “Determination of yield stress of 2D (Yukawa) dusty plasma.” Two major findings were reported in the paper. In the low-force elastic limit, a measurement of the shear modulus is obtained, which agrees with a theoretical value based on a sound speed. For larger forces resulting in plastic deformation, we determine the yield stress, which is found to agree with a common theoretical model for the critical yield stress.

	We also performed another simulation to support our breadboard experiments. This work was to test a fluctuation theorem in statistical physics. Our simulation results agree with the results from the breadboard experiments. This simulation effort contributed to a paper published in Nature Physics. BROADER IMPACTS A K-12 outreach presentation was made at an elementary school. Our research results have been disseminated in journals and at conferences. Five papers have been published in peer-reviewed journals. Two invited and seven contributed talks were presented at international conferences. We also distributed to other PIs a copy of software that we developed at Iowa.
Bibliography Type:	Description: (Last Updated: 10/05/2022)
Abstracts for Journals and Proceedings	Goree J. "Statistical physics experiments using dusty plasmas." Presented at the 8th International Conference on the Physics of Dusty Plasmas, Prague, Czech Republic, May 20-25, 2017. 8th International Conference on the Physics of Dusty Plasmas: Book of Abstracts: Prague, May 20-25, 2017. , May-2017
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Abstracts for Journals and Proceedings	Liu B, Goree J, Pustylnik M, Thomas H, Fink M, Morfill G, Fortov V, Usachev A, Molotkov V, Lipaev A, Petrov O, Thoma M. "Particle velocity distribution in a three-dimensional dusty plasmas." Presented at the 8th International Conference on the Physics of Dusty Plasmas, Prague, Czech Republic, May 2017. 8th International Conference on the Physics of Dusty Plasmas: Book of Abstracts: Prague, May 20-25, 2017. , May-2017
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